EXPERIMENT-07

AIM: To write a program to implement flow at data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow.

CODE:-

SENDER.PY

```
import time
w = int(input("Enter Window Size:"))
msg = input("Enter Message:")
frames = [[i, ch] for i, ch in enumerate(msg)]
base = 0
while base < len(frames):
    with open("Sender_Buffer.txt", "w") as f:
        for i in range(base, min(base+w, len(frames))):
            f.write(f"{frames[i][0]}{frames[i][1]}\n")
print("In Frames Sent:")
for i in range(base, (min(base+w), len(frames)):
    print(frames[i])
time.sleep(3)
try:
    with open("Receiver_Buffer.txt", "r") as f:
        data = f.read().strip()
    if data.startswith("Ack"):
        ack = int(data.split(":")[1])
        if ack >= base+w-1 or ack >= len(frames):
            print("Acks received:", ack)
            base = ack+1
        else:
            print("Wrong Ack Resending...")
```

```
elif data.startswith("NAck"):
        print("NAck Recieved. Resending ...")
except:
    print("Waiting for Reciever")
```

RECIEVER.PY

```
exp = 0
with open("Sender_Buffer.txt","r") as f:
        lines = f.readlines()

ok = True
for line in lines:
    no, data = lines.strip().split(";")
    if int(no) != exp:
    ok = false
    nack = exp
    break
  exp += 1
with open("Reciever_Buffer_txt","w") as f:
  if ok:
    f.write(f"Ack: {exp-1}")
    print("Ack Sent:", exp-1)

  else:
    f.write(f"NACK: {nack}")
    print("NACk Sent:", nack)
```

INPUT:

Enter Window Size: 3

Enter Message: HELLO

OUTPUT:-

Frames sent:

[0, 'H']

[1, 'E']

[2, 'L']

Ack receiver: 2

Frames sent:

[3, 'L']

[4, 'O']

Ack receiver: 4

OUTPUT:

Ack sent: 2

Ack sent: 4

Result:

Thus, the program to Implementation flow control at data link layer used SLIDING WINDOW has been executed successfully.